

MATH 225 – Applied Linear Algebra II

Course Notes

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Contents

1	Chapter 1: General Vector Spaces	2
1.1	The elements of a vector space V are Called vectors. The elements of the field $\mathbb{R}$ are called scalars.	2

Chapter 1: General Vector Spaces

Vector Addition Axioms

1. **Closure under Addition:** For all $u, v \in V$, $u + v \in V$.
2. **Commutativity:** $u + v = v + u$ for all $u, v \in V$.
3. **Associativity:** $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$.
4. **Additive Identity:** There exists $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
5. **Additive Inverse:** For each $v \in V$, there exists $-v \in V$ such that $v + (-v) = 0$.

Scalar Multiplication Axioms

6. **Closure under Scalar Multiplication:** For all $c \in \mathbb{R}$ and $v \in V$, $cv \in V$.
7. **Vector Distributivity:** $c(u + v) = cu + cv$ for all $c \in \mathbb{R}$ and $u, v \in V$.
8. **Scalar Distributivity:** $(c + d)v = cv + dv$ for all $c, d \in \mathbb{R}$ and $v \in V$.
9. **Associativity:** $(cd)v = c(dv)$ for all $c, d \in \mathbb{R}$ and $v \in V$.
10. **Multiplicative Identity:** $1v = v$ for all $v \in V$.

The elements of a vector space V are Called vectors. The elements of the field $\mathbb{R}$ are called scalars.

Example of V.S.

$$(1) \mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{R} \right\} \text{ under standard Vector + Scalar Multiplication}$$

$$(2) M_{m \times n}(\mathbb{R})$$

$$(3) P_n(\mathbb{R}), \text{ space of all polynomials with degree } \leq n$$

Define as follows: $(f + g)(x) = f(x) + g(x)$

“Scalar Multi”: $(cf)(x) = cf(x) \quad (c \in \mathbb{R})$

Then $P_n(\mathbb{R})$ with + Scalar Multiplication, is a v.s over $\mathbb{R}$